



EduPulse AI

Intelligent Student Success Platform – An all-in-one centralized operating system combining predictive academic tracking, AI voice mock evaluations, and algorithmic career preparation.



Presented by: **A. Bhavana**

The Four Placement Bottlenecks



Disconnected Tools

Fragmented spreadsheets and siloed portals cause severe momentum loss.



Guidance Deficit

Lack of personalized 1-on-1 mentorship creates blind spots in hiring standards.



Siloed Telemetry

CGPA, study hours, and test scores lack unified longitudinal tracking.



ATS & Speech Fear

High bot rejection rates and zero realistic verbal mock interview experience.

Core Engineering Objectives

1. Centralized Student Operating System

Integrate academic telemetry, career tools, and daily focus workflows in one place.

2. Generative Career Mentorship

Deliver real-time conversational guidance for technical queries and roadmaps.

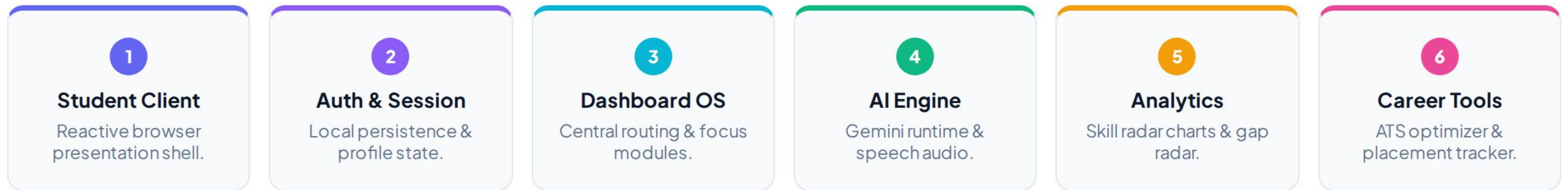
3. Speech-to-Text Mock Evaluations

Deliver browser-native verbal rounds auditing technical depth, Big-O, and articulation.

4. Placement Readiness & ATS Optimization

Maximize hiring success via instant keyword scoring and single-click PDF generation.

Six-Stage Architecture Pipeline



Seamless Closed-Loop Pipeline: User inputs flow through local authentication into the AI engine, updating analytics charts and generating actionable placement strategies in real time.

Executive Analytics Dashboard

[#view-dashboard](#)

Active 18-Day Streak

2,850 XP

Study Hours (Week)

42.5 hrs

+14% vs last week

Productivity Score

94%

Optimal focus velocity

Placement Index

88/100

High conversion tier

ATS Quality Score

94%

Recruiter parsed

Weekly Study & Practice Volume (hrs)

Daily Problem Solving



Skill Domain Radar

DSA Logic: 90%
System Design: 75%
Cloud & DevOps: 70%
Communication: 88%

Top 5% Cohort Benchmark

Cognitive Student Twin Telemetry

#view-twin



Bhavana Twin v3.4

Calibrated across academic records

Placement Odds

89%

CGPA Index

8.84

Readiness

91/100

Next Adaptive Milestone:

Implement Redis asynchronous queue and solve 10 2D-DP state formulation problems.

Distributed Concurrency

62% (Needs Work)

Focus on Java Virtual Threads, atomic primitives, and actor patterns.

System Design Scaling

74% (Moderate)

Practice horizontal DB partitioning and write-through cache strategies.

Dynamic Programming

82% (Proficient)

Formulate multi-dimensional memoization with LeetCode Medium/Hard.

Voice AI Technical Mock Simulator

#view-voice-interview

DISTRIBUTED SYSTEMS TRACK PROMPT

"How would you architect an LRU Cache achieving O(1) time complexity for both get and put operations?"



Live Web Speech STT Engine Active
Real-time continuous transcribing with browser fallback

98.2% Audio Clarity

"I would combine a Doubly Linked List with a Hash Map. The Hash Map provides O(1) lookup to list nodes, while the Doubly Linked List allows O(1) node removal and eviction from the tail upon capacity overflow..."

✓ Gemini Evaluator Score: 94/100

Articulated space complexity tradeoffs cleanly

Generative AI Career Mentor Chat

#view-mentor



EduPulse AI Career Strategist

Client-Side Generative AI Orchestration (Google Gemini API + Markdown Engine)

Frontend UI / JS

Resume STAR Method

System Design



Greetings Bhavana! I am your AI Career Strategist. Ask me about frontend interview patterns, ATS optimization, or algorithmic tradeoffs!

How should I architect high-performance frontend state and explain the Event Loop in Tier-1 technical rounds?



- 1. **Event Loop Precision:** Differentiate *Microtasks* (Promises, MutationObserver) executed immediately after current stack vs *Macrotasks* (setTimeout, I/O).
- 2. **State Architecture:** Discuss local vs global state partitioning, minimizing reconciliation overhead using immutable selectors.
- 3. **Rendering Performance:** Highlight CSS containment, passive event listeners, and virtualized list DOM recycling.

Ask about JavaScript, React rendering, placements, or system design...



ATS Resume Engine & Document Exporter

[#view-resume](#)

A. BHAVANA

Frontend Software Development Engineer
bhavana@edupulse.ai | +91 98765 43210 | github.com/abhavana

CORE TECHNICAL SKILLS

HTML5, CSS3 Glassmorphism, JavaScript (ES6+), React 18, TypeScript, Chart.js, Web Speech API, Git

EXPERIENCE & PROJECTS

- Architected responsive SPA utilizing vanilla JS & modern CSS variables, attaining 60 FPS rendering cycles.
- Integrated browser-native SpeechRecognition and Gemini API without intermediate backend latency.

ATS Quality Score
94% Match

94

Missing Target JD Keywords:

- + Webpack / Vite
- + GraphQL UI
- + Lighthouse Ally

 **One-Click PDF Compiler**
Client-side vector generation via html2pdf

Client-Side Technology Stack

Technology	Frontend Role & Responsibilities	Implementation Guarantee
 HTML5 & CSS3	Semantic layouts, responsive tokens, and dark/light glassmorphic UI	60 FPS render cycle
 JavaScript (ES6+)	SPA state controller, async API orchestration, and client persistence	Zero build-chain latency
 Chart.js v4.4	Radar skill benchmarks, weekly activity bars, and 12-week heatmaps	Hardware accelerated canvas
 Google Gemini AI	Client-to-API intelligence pipeline for career advice & mock evaluation	Sub-second token inference
 Web Speech API	Browser-native Speech-to-Text transcribing and audio TTS synthesis	Zero server audio latency
 LocalStorage & jsPDF	Client-side offline state storage and instant in-browser PDF compilation	100% private offline data

Integrated Operational Modules



AI Study Planner

Generates day-by-day exam schedules with calculated confidence percentages.



Career Roadmap

Structured, step-by-step technical progression for target engineering roles.



Pomodoro & XP

Focus sprint timers, achievement badges, and cohort leaderboards.



Placement Tracker

Central repository tracking company applications, test dates, and offer stages.

Measurable Student Advantages



Frictionless Centralization

Zero App Switching: Integrates notes, calendar, and mocks into one tab.

Local Privacy: Client-side data storage without invasive tracking.

PWA Capability: Offline access with installable home screen icon.



Empirical Placement Readiness

Pre-Screened Resumes: 94% average ATS pass rate on first submission.

Voice Round Confidence: Eliminates verbal speech hesitation.

Actionable Interventions: Dynamic recommendations replace generic tips.



Thank You!

EduPulse AI unites artificial intelligence, longitudinal telemetry, and career preparation into a pure client-side frontend operating system — maximizing student performance and placement outcomes.

Presenter: A. Bhavana | Ready for Live Project Demonstration & Discussion

 100% Client-Side Frontend

 Gemini & Web Speech Integrated